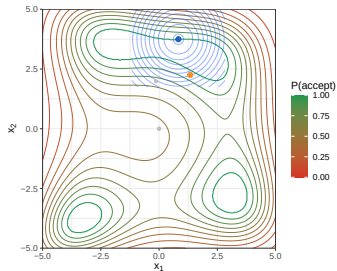
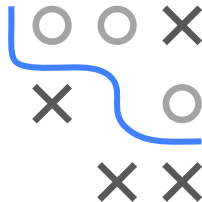


# Optimization in Machine Learning

## Simulated Annealing

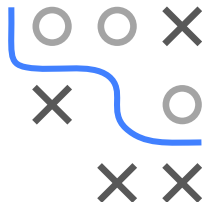


### Learning goals

- Motivation
- Metropolis algorithm
- Simulated Annealing

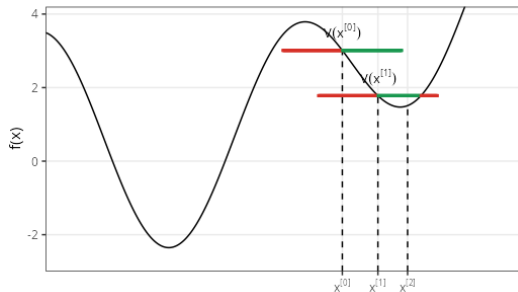
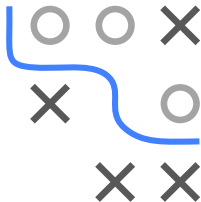
# INTRODUCTION

- Simulated annealing: heuristic for optimization of complex objectives  
(multivariate, non-linear, non-convex)
- Does not guarantee global optimum, but usually good solutions
- Goal for such problems: avoid “getting stuck” in local optima
- Often used for difficult discrete problems as well
- Local search strategy with random option to accept worse values, based on the Metropolis algorithm (see below)



# SIMPLE STOCHASTIC LOCAL SEARCH

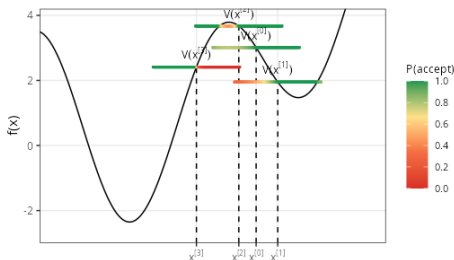
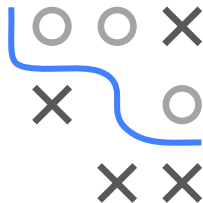
- Given is a multivariate objective function  $f(\mathbf{x})$
- Define a local neighborhood area  $V(\mathbf{x})$  for a given  $\mathbf{x}$
- Sample proposal  $\mathbf{x}^{[t+1]}$  uniformly at random from  $V(\mathbf{x}^{[t]})$
- Calculate  $f(\mathbf{x}^{[t+1]})$
- If  $\Delta f = f(\mathbf{x}^{[t+1]}) - f(\mathbf{x}^{[t]}) < 0$ ,  $\mathbf{x}^{[t+1]}$  is accepted as new solution, otherwise a new proposal from neighborhood is sampled



Stoch. local search: acceptance range in green and rejection in red

# METROPOLIS ALGORITHM

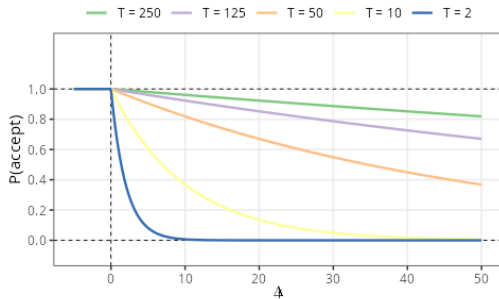
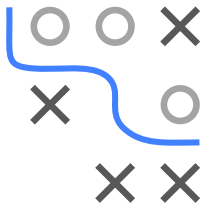
- Simple stochastic local search strongly depends on  $\mathbf{x}^{[0]}$  and the neighborhood  
⇒ Danger of ending up in local minima
- **Idea:** allow worse candidates with some probability
- **Metropolis:** accept candidates from previous rejection range ( $\Delta f > 0$ ) with probability  $\mathbb{P}(\text{accept} \mid \Delta f) = \exp(-\Delta f / T)$
- $T$  denotes “temperature”



Simulated annealing: colors correspond to  $\mathbb{P}(\text{accept})$

# METROPOLIS ALGORITHM

- High temperatures correspond to high probability of accepting worse  $x$
- Local minima can be escaped, but no convergence can be achieved at *constant* temperature
- We come across an important principle of optimization:  
**exploration (high T) vs. exploitation (low T)**



# SIMULATED ANNEALING

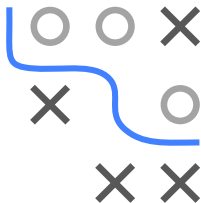
- Start with high temperature to explore whole space
- Slowly reduce temperature to converge  
⇒ Sequence of descending temperatures  $T^{[t]}, t \in \mathbb{N}$
- Procedure is called simulated annealing
- Temperature is often kept constant several iterations in a row to explore the space, then multiplied by coefficient  $0 < c < 1$ :

$$T^{[t+1]} = c \cdot T^{[t]}$$

- Other strategies possible, for example:

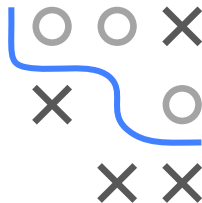
$$T^{[t]} = T^{[0]} \left( 1 - \frac{t}{t_{\max}} \right)$$

- Also many different strategies for choosing the neighborhood, strongly depending on the objective function



# ANALOGY TO METALLURGY

- Simulated annealing draws an analogy between a cooling process (e.g., of a metal or liquid) and an optimization problem
- Consider the atoms of the liquid as a system with many degrees of freedom
- If the liquid is cooled too fast, it solidifies in a suboptimal configuration, whereas slow cooling produces crystals with optimal structure (minimum energy state)
- The minimum energy state corresponds to the optimum of the objective function



# PROS AND CONS

## Pros:

- Very cheap in terms of operations per iteration
- “Meta” heuristic: works even for complicated or discrete problems

## Cons:

- Many control parameters; how exactly to cool down and sample is unclear (countless variants exist)

